

Sudanese

Senna pods

قرون السنمكة السودانية

Senna alexandrina

Sun-dried and cleaned



<i>Senna alexandrina</i> Mill.	Alexandrian senna	1211.90
Botanical source	Type	HS code (indicative)

1 Scope

This sheet specifies the conditions and characteristics of senna pods exported by Asmagh Sudanese Exports from Port Sudan.

2 Synonyms

Alexandrian senna pods

Sennae fructus

Cassia senna L.

Cassia acutifolia Delile

3 Definition

The dried ripe fruit of *Senna alexandrina* Mill. (syn. *Cassia senna* L.), family Fabaceae (Leguminosae).

4 Description

Flat, thin, leaf-like pods, oblong or kidney-shaped, pale to greyish green or yellowish brown with a darker area over the seeds; 3.5–6.0 cm long and 1.4–1.8 cm wide, with the remains of the style at one end. Each pod holds 6–10 obovate, green to pale brown seeds with prominent lengthwise ridges on the seed coat.

5 Grade and forms

Clean	Sun-dried and cleaned pods
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Available as

Whole pods

Powder

6 Uses



Herbal teas



Pharmaceutical sennoside extracts



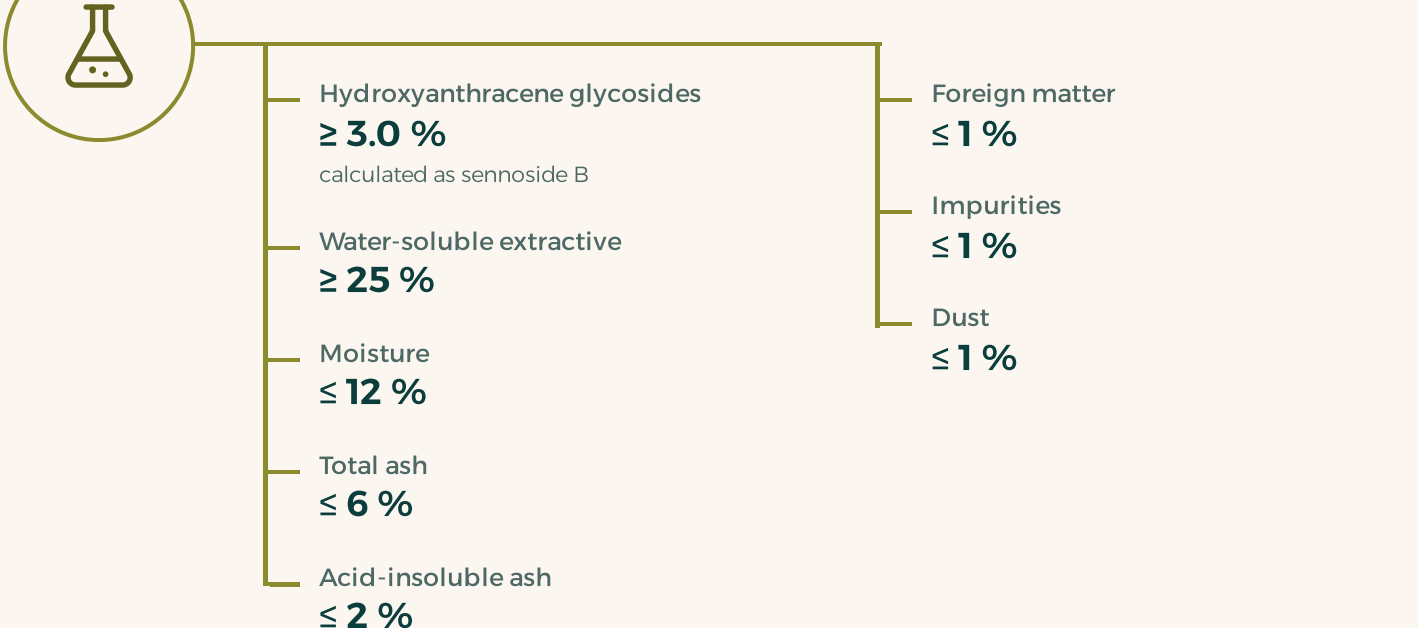
Powder for formulation

7 Requirements

7.1 General conditions

- ✓ Free from lumps and mould
- ✓ Free from live or dead insects, their parts and rodent remains
- ✓ Free from fertilisers and harmful chemicals; pesticide residues within the limits in section 7.4
- ✓ Free from foreign odours and off-odours
- ✓ Free from polluting substances and toxins such as aflatoxins

7.2 Measurement requirements



Percent by mass. Water-soluble extractive, moisture, ash and foreign matter limits match the WHO monograph for senna fruit (Fructus Sennae).

7.3 Microbiological limits

Aerobic bacteria	≤10 ⁵ cfu/g	Other enterobacteria	≤10 ³ cfu/g
Moulds and yeasts	≤10 ³ cfu/g	<i>Escherichia coli</i>	Absent
<i>Salmonella</i> spp.	Absent		

7.4 Contaminants

Heavy metals	Lead ≤10 mg/kg and cadmium ≤0.3 mg/kg, as recommended by WHO for the final dosage form
Pesticide residues	Aldrin and dieldrin ≤0.05 mg/kg; other pesticides within national limits and WHO guidelines
Radioactive residues	Tested according to the WHO guidelines for quality control of medicinal plant materials

7.5 For pharmaceutical buyers

Minimum hydroxyanthracene glycosides, calculated as sennoside B on the dried drug:

European Pharmacopoeia	≥ 3.4 % for Alexandrian pods; ≥ 2.2 % for Tinnevely pods
WHO monograph (Fructus Sennae)	≥ 2.2 %

Our minimum is 3.0 %. The European Pharmacopoeia listed separate Alexandrian and Tinnevely pod monographs up to its 9th edition; check the edition your buyer uses, and ask us for lot results before contracting to a pharmacopoeia grade.

8 Microscopy

Under the microscope, the pod shows (WHO monograph):

Epicarp

Very thick-cuticled, isodiametric cells with occasional anomocytic or paracytic stomata and very few unicellular, warty hairs.

Hypodermis and mesocarp

Hypodermis of collenchyma; mesocarp of parenchyma containing a layer of calcium oxalate prisms.

Endocarp

Thick-walled fibres, mostly at right angles to the long axis of the fruit; the inner fibres run obliquely or parallel to it.

Seeds

A layer of palisade cells with thick outer walls under the seed coat; endosperm of polyhedral cells with mucilaginous walls.

9 Packaging

Packed in approved, clean, dry and well-closed packaging that protects the pods from light and moisture, keeps their properties, and neither affects the product nor is affected by it. Asmagh ships in pressed bales of 200–250 kg.

Container load about 19 MT per 40 ft container.

10 Storage

Stored in a clean place, low in humidity and heat, and protected from light and dust.



11 Labelling

Every package carries a clear identifying label showing:

Name of the crop  Senna pods	Type, exporter or trade mark  Asmagh Sudanese Exports	Season of production  Season and year	Code  Lot code
Country of origin  Sudan	Place of processing  Sudan	Net weight (kg)  Per bale, 200–250 kg	Writing  Clear and permanent

12 Shelf life

No more than two years from the date of production.

13 Sampling and testing

Samples are taken according to the Sudanese sampling standard No. 307. Hydroxyanthracene glycosides are determined on that sample and calculated as sennoside B by the British Pharmacopoeia method.

Each shipment carries an analysis and quality certificate issued by SSMO.

From field to port

من الحقل إلى الميناء

Every lot is sampled on arrival, sun-dried, cleaned and tested before it is baled. Storage and loading are checked too, because quality can be lost after processing.

1 Our quality process

1



Sourcing

Bought from growers and collectors in northern Sudan; every incoming lot is checked.

2



Sun-drying

Pods dried in the sun, away from dust and contamination.

3



Cleaning

Stems, leaves, sand and foreign matter removed.

4



Testing

Samples tested for sennosides, moisture, ash and microbiology.

5



Baling

Pressed into 200–250 kg bales marked with product, lot and weight.

6



Storage

Kept dry, cool and out of the light until loading.

7



Loading

Containers inspected before stuffing; loading is supervised.

8



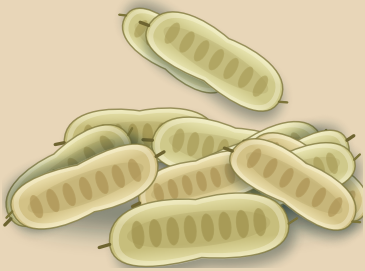
Shipping

Shipped from Port Sudan with the full set of export documents.

2 Origin and shipping

Senna grows wild and under cultivation in northern Sudan, along the Nile, and has been traded as a medicinal herb for centuries. The dried pods travel to Port Sudan on the Red Sea for export.

HS code	1211.90 (indicative; confirm with your customs broker)
Port of loading	Port Sudan (SDPZU), Red Sea
Container load	About 19 MT per 40 ft container
Ready to load	Usually 10–15 days after contract
Transit	About 2–8 weeks, by destination
Delivery terms	FOB Port Sudan, CFR or CIF
Payment terms	Irrevocable L/C at sight, TT advance, or D/P
Samples	Free; courier paid by the buyer



3 Export documents

Certificates travel with each shipment and are exchanged through the banks or directly with the buyer, so the goods clear customs at destination.

Product certificates

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Certificate of origin
Sudan Chamber of Commerce; general or preferential forms.
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Phytosanitary certificate
Official inspection against the importing country's pest list.
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Analysis and quality certificate
Issued by the Sudanese Standards & Metrology Organization (SSMO).
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Fumigation certificate
Treatment, chemical, dose and temperature used.
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Inspection certificate
Optional; by an international inspection company chosen by the buyer.

Shipping documents

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Export contract
Attested by the Ministry of Trade and the Chamber of Commerce.
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Commercial invoice
Goods, quantity, value and terms.
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Packing list
Every bag in each container, with counts and weights.
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Bill of lading
Issued by the shipping line on loading.

4 Standards and references

- World Health Organization. WHO monographs on selected medicinal plants, Volume 1: Fructus Sennae. Geneva, 1999.
- World Health Organization. Quality control methods for medicinal plant materials. Geneva, 1998.
- European Pharmacopoeia: monographs Senna pods, Alexandrian (0207) and Senna pods, Tinnevely (0208).
- British Pharmacopoeia (2000); British Herbal Pharmacopoeia (1983, 1996).
- Youngken, H. W. Textbook of Pharmacognosy, 6th edition. Blakiston, Philadelphia, 1950.

